

RMA Process Improvements



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LEAVEY SCHOOL OF BUSINESS

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I. Project Introduction

INTRODUCTION

Scope and Objectives

Review one year of historical RMA data to identify enhancements for driving process efficiency and cost savings.

- **Analyze Engineer-Customer Alignment:** Evaluate the alignment between engineer-provided resolution summaries and customer-provided problem details using similarity scores to uncover actionable insights for driving process improvement initiatives.
- **RMA Analysis for Automation:** Analyze past RMA cases to identify recurring themes and common issues, enabling the prioritization of automation initiatives to streamline operations and enhance service delivery.
- **Analyze Seasonal Demand Trends:** Analyze RMA data to identify seasonal demand patterns, enabling strategic decision-making and targeted automation initiatives to optimize resource allocation, enhance efficiency, and ensure timely customer responsiveness.

INTRODUCTION (Cont'd)

Major Work Activities

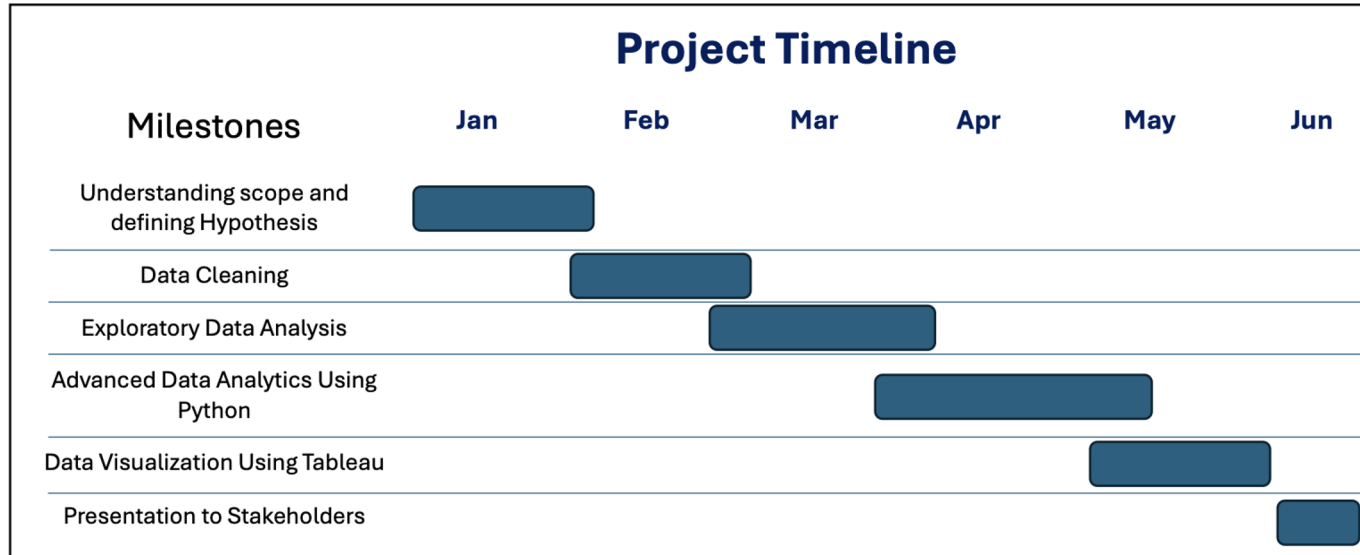
- Conduct data cleaning to handle missing values, remove duplicates, and correct any inconsistencies to ensure accuracy in the analysis.
- Review and analyze approximately 290k data points spanning from July 2022 to July 2023.
- Calculate cosine similarity to measure how closely the summaries of engineer solutions ('RESOLUTION_SUMMARY') match the customer descriptions of problems ('PROBLEM_DETAILS').
- Identify specific RMAs that are suitable for automation based on frequency, complexity, and the accuracy of past cases.
- Analyze trends over the year to identify any seasonal patterns or anomalies in RMA occurrences.
- Apply NLP techniques to analyze textual data from columns like PROBLEM_DESCRIPTIONS to identify recurring topics in the textual data to uncover patterns and common issues.

INTRODUCTION (Cont'd)

Summary of Analysis

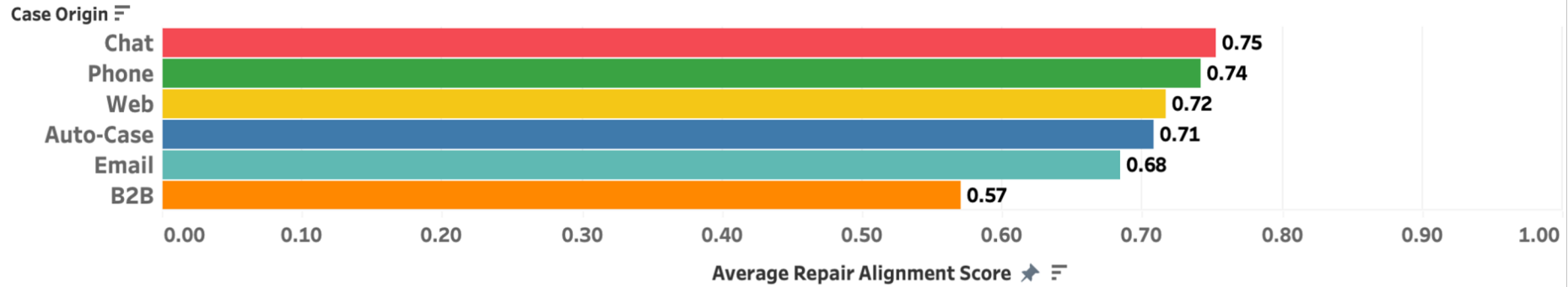
- **Repair Alignment:** Utilize cosine similarity to measure the alignment between customer-reported issues and engineering solutions, identifying candidates for automation.
- **SMB Channel:** Analyze RMA request data from different channels to understand submission patterns and resolution times, with a focus on optimizing processes for SMB clients.
- **Educational Resolution:** Investigate the impact of customer education on support case volume, exploring ways to reduce cases through improved educational resources.
- **RMA Approval:** Examine RMA approval rates across different problem details to identify trends and areas for improvement in RMA approval processes.
- **Seasonality and Cost:** Explore seasonal trends in RMA occurrences and associated costs across various product families to inform resource allocation and planning decisions.
- **Product Families:** Analyze issue frequency and resolution times across different product families to identify areas for process improvement and the potential benefits of standardizing resolutions.

INTRODUCTION (Cont'd)



II. Repair Alignment

Chat and Phone channels show a high repair alignment score between customer reporting and engineering response.

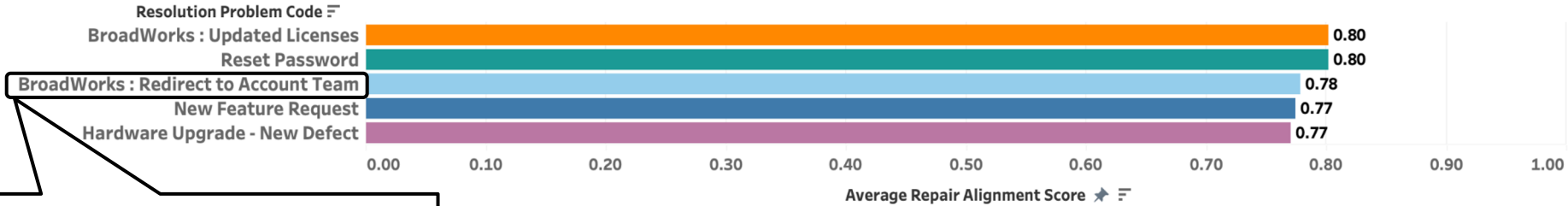


Note

- *Repair Alignment: Measured using cosine similarity scores that determine how closely the solutions match the problem descriptions, calculated through techniques in Natural Language Processing (NLP).*
- *The average repair alignment score for the entire dataset is **0.72**. Scores above the average indicate candidates for automation.*

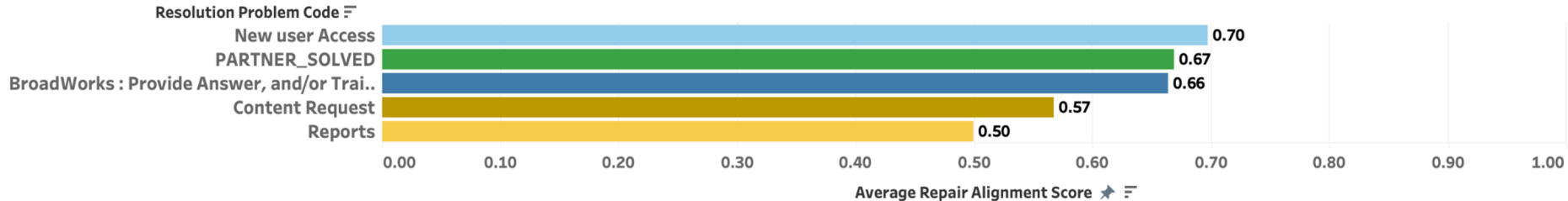
High repair alignment scores for Licensing and passwords indicate that associated RMA tickets should be automated.

Top 5 scores



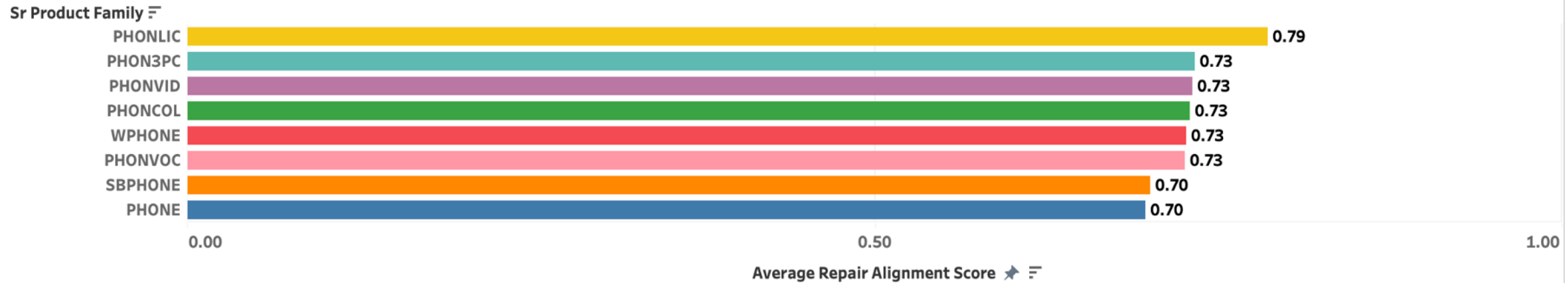
Non-RMA inquiries should be redirected to the appropriate team automatically by the system.

Lowest 5 scores



**RMAs for PHON product families should be automated,
given above-average repair alignment scores and low repair cost*.**

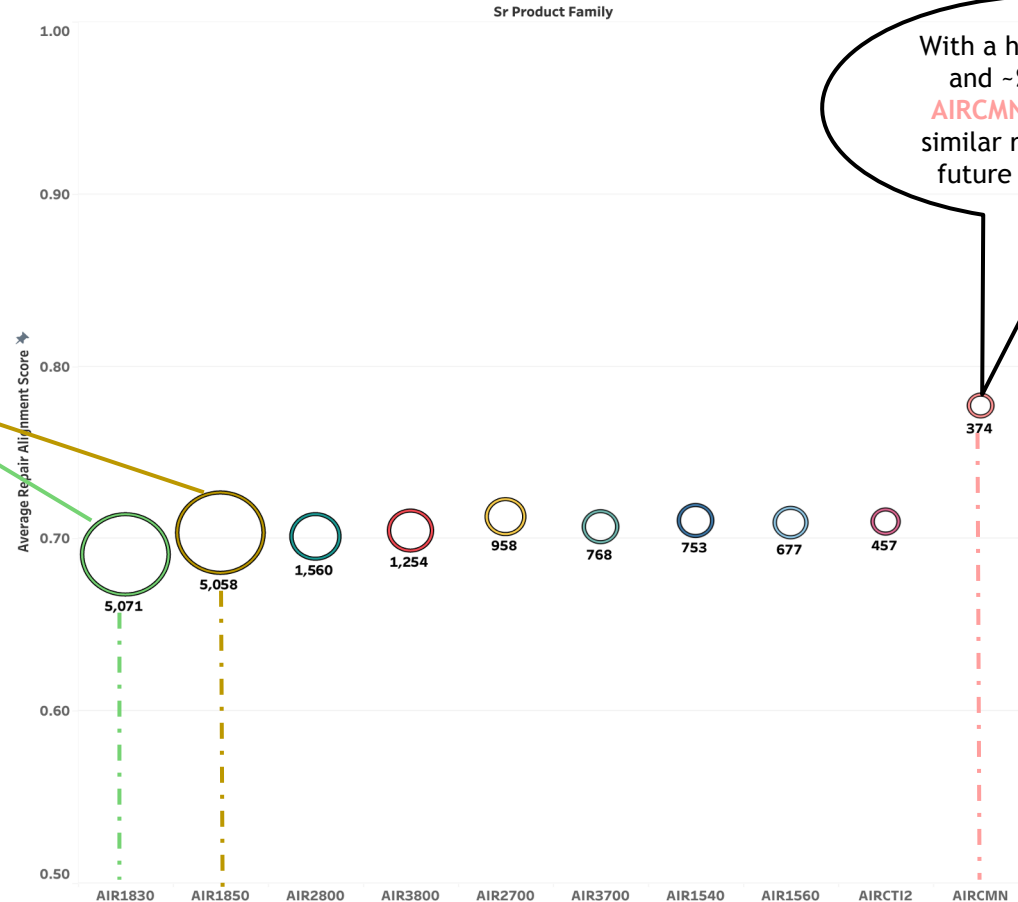
PHON



* ~80 USD unit cost

Reduce warranty coverage for certain EOL products to encourage customers to transition to the latest product generation.

Top 10 Count for the AIR Family by Repair Alignment Scores



Aironet 1830 and 1850 reached their EOL in November 2022, with scores of 0.69 and 0.70 respectively.

With a high score 0.78 and ~\$80 cost for AIRCMN, automating similar requests in the future is suggested.

Note

The size of each circle reflects the count of occurrences.



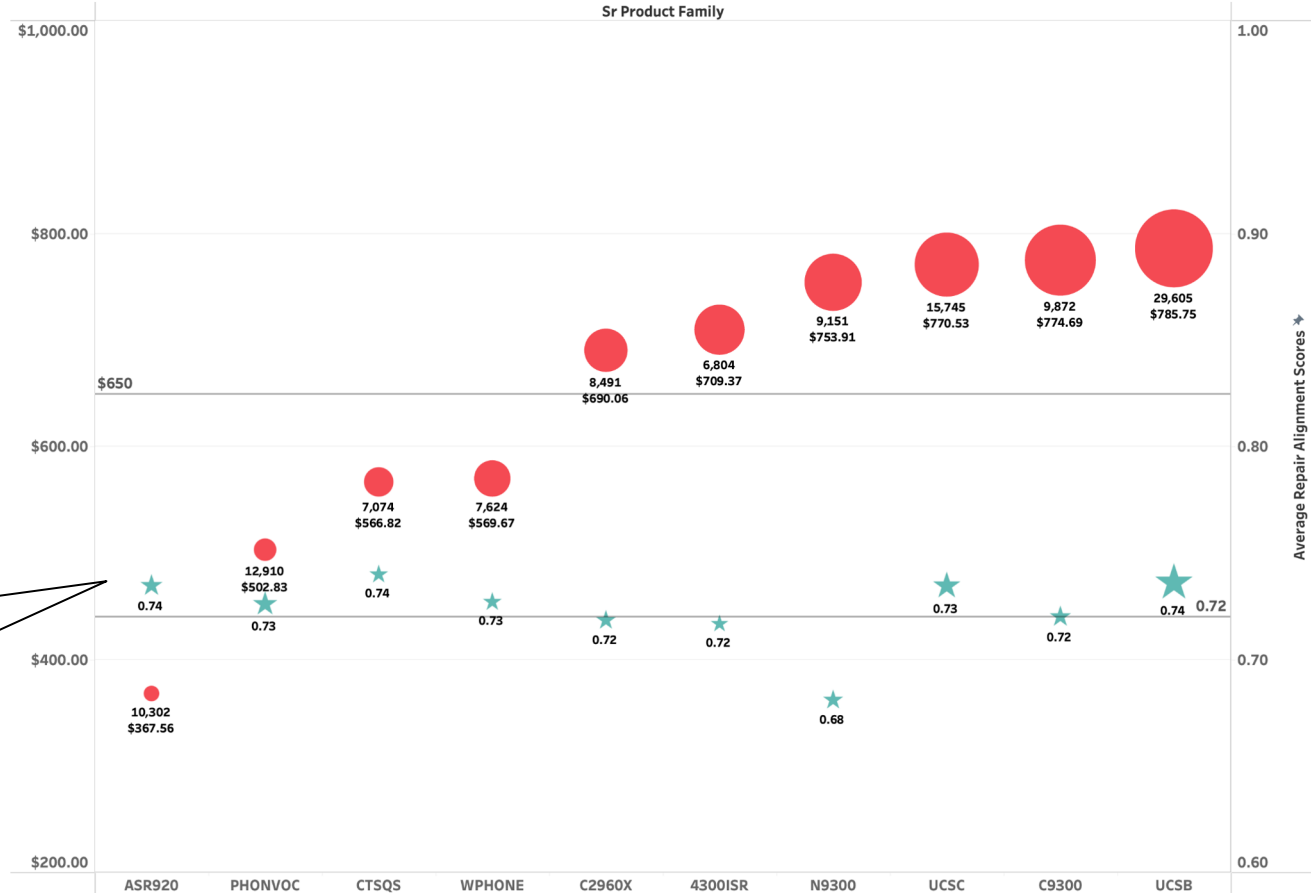
Review RMAs for automation that have high volume, median allocated cost below \$654, and repair alignment score greater than 0.72.

Median Allocated Cost and Repair Alignment Score by Product Families

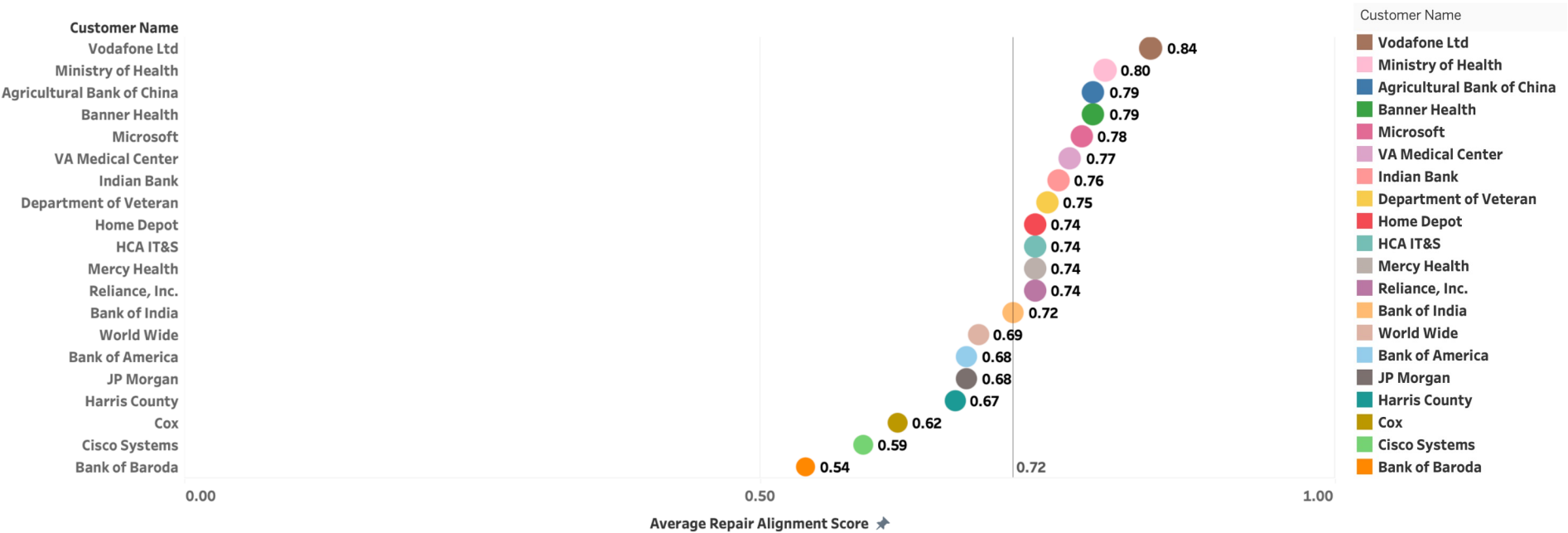
Note

- The median allocated cost for the entire dataset is **\$654**.
- The size of each circle reflects the count of occurrences.

ASR920, PHONVOC, CTSQS, WPHONE



Customers with repair alignment scores above 0.72 are candidates for future automation.



Note

These are the top 20 customers in 2022.

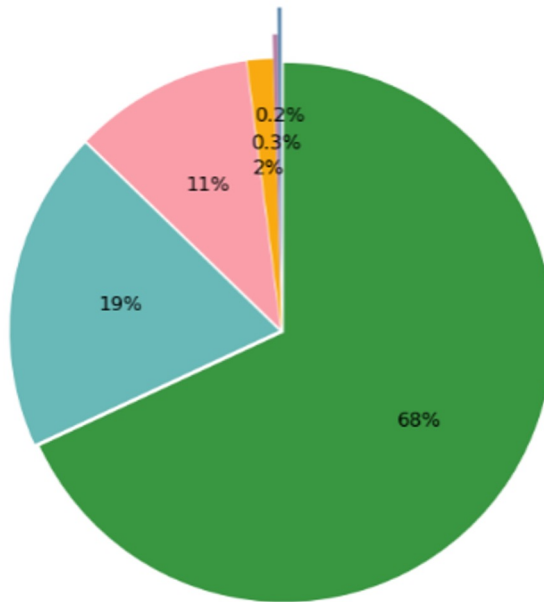
Recommendations

- 1) The Web channel should be prioritized due to its low cost and the ability to provide more structured data.
- 2) Phone products with repair alignment scores above 0.72 and costs below \$100 should be automated.
- 3) Discontinue or reduce warranty coverage for EOL products with high volume and high repair alignment scores to encourage customers to transition to the next product generation.
- 4) Review RMAs for automation for major customers with repair alignment scores of 0.72 or higher.
- 5) To better classify RMA issues we suggest the following:
 - a) Enhance data input quality by refining and expanding problem codes, thereby allowing customers to select codes that more accurately reflect their issues and ensuring more precise data collection.
 - b) Encourage engineers to provide detailed descriptions of the repairs made.

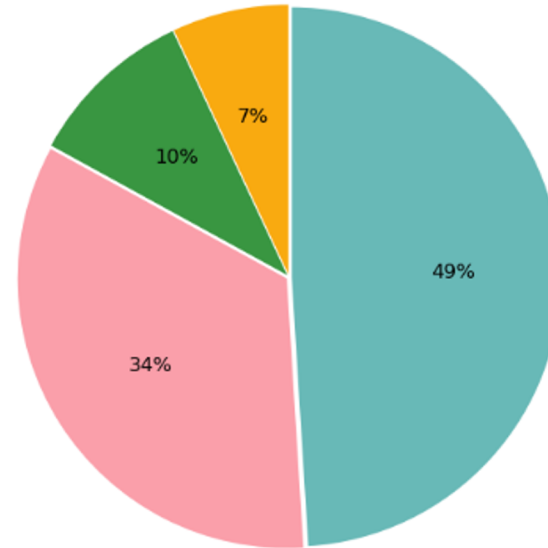
III. SMB Channel

Nearly half of SMB RMA requests initiated through phone channels.

Total Cases by Case Origin

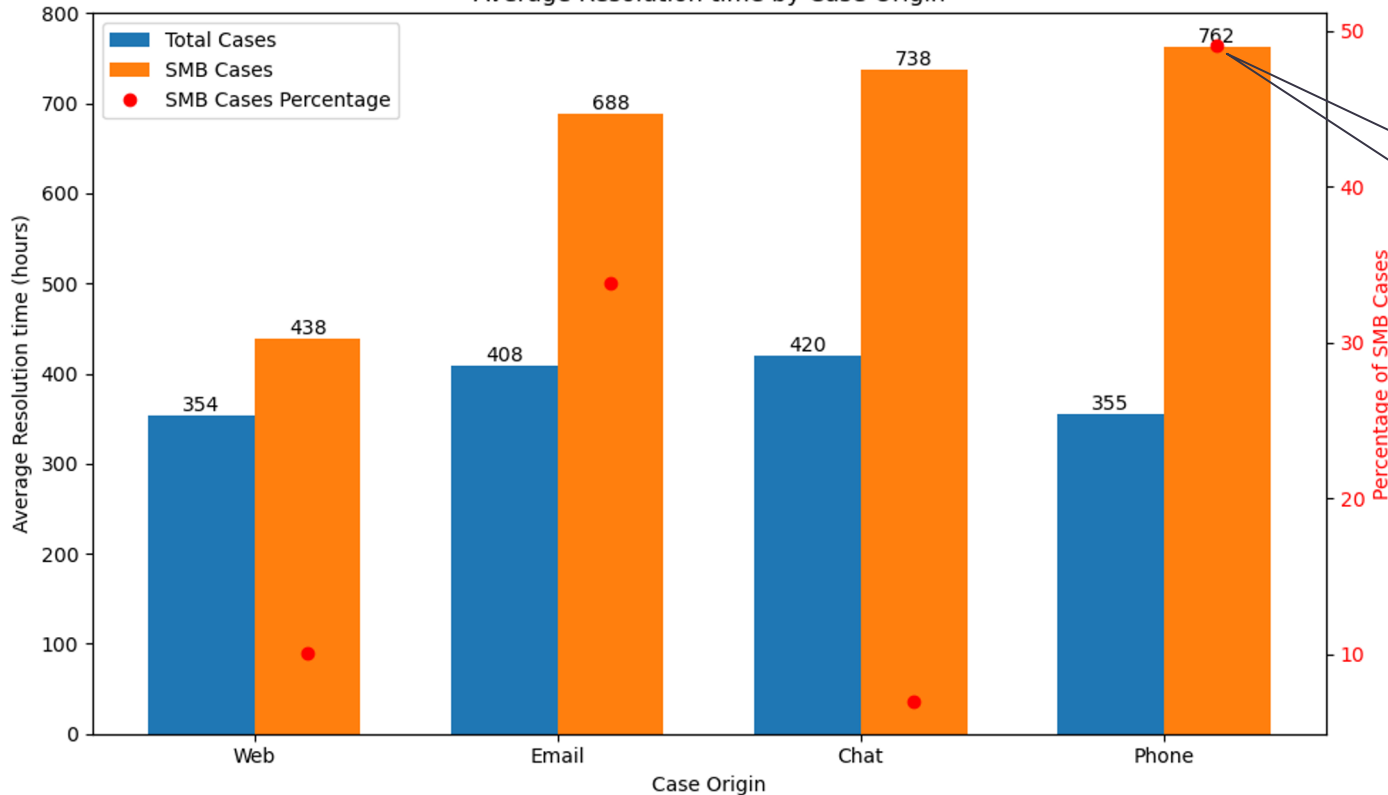


SMB Cases by Case Origin



SMB cases require twice the resolution time of other segments.

Average Resolution time by Case Origin

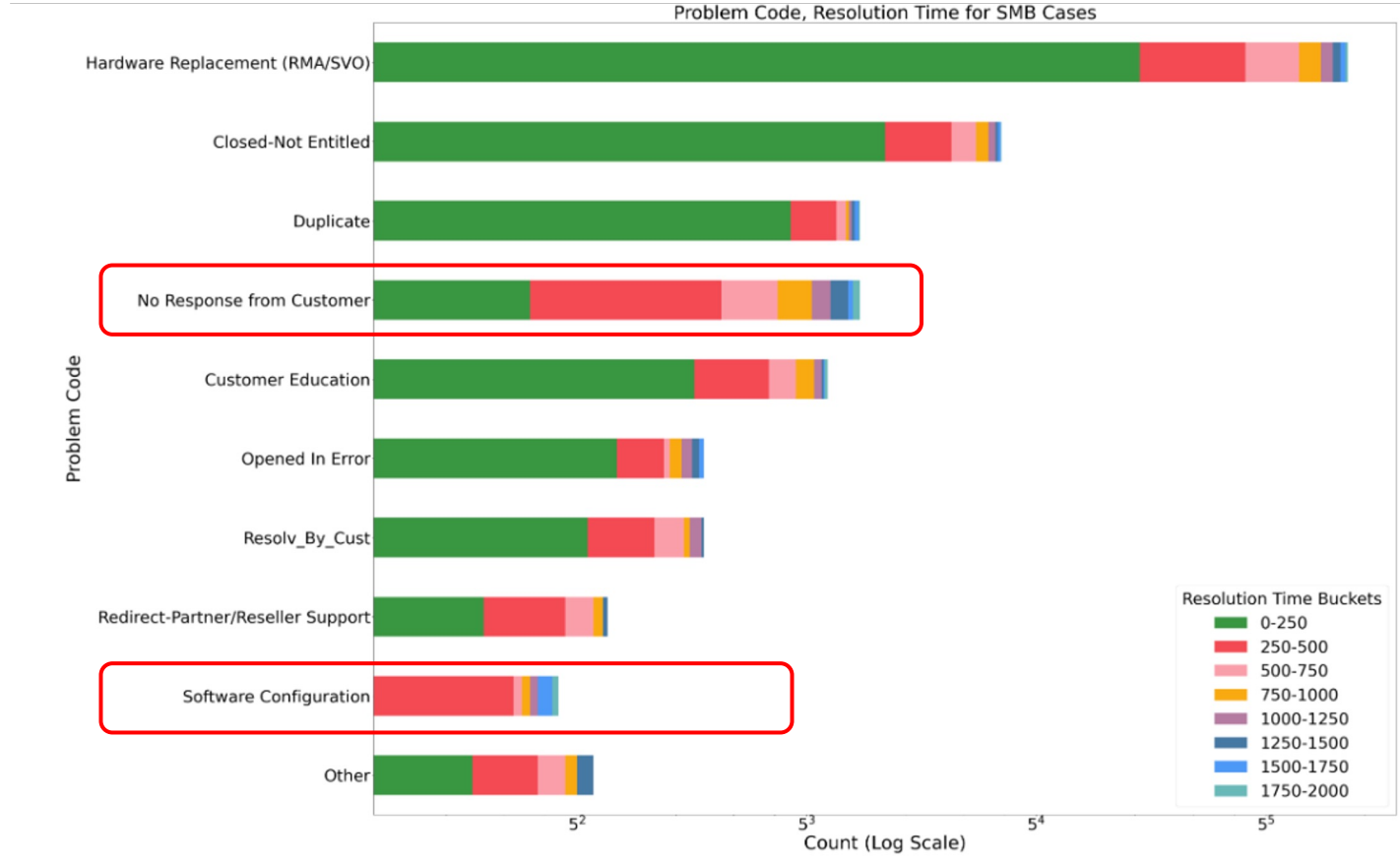


Resolving SMB cases takes twice as long compared to other segments on the costlier channel .

Promote and incentivize the use of web-based RMA submission for SMB clients, emphasizing its convenience and efficiency over traditional phone channels. This would drive down operational expenses, minimize wait times, and enhance the scalability of request management.



“Software Configuration” and “No Response from Customer” show significantly longer resolution time.



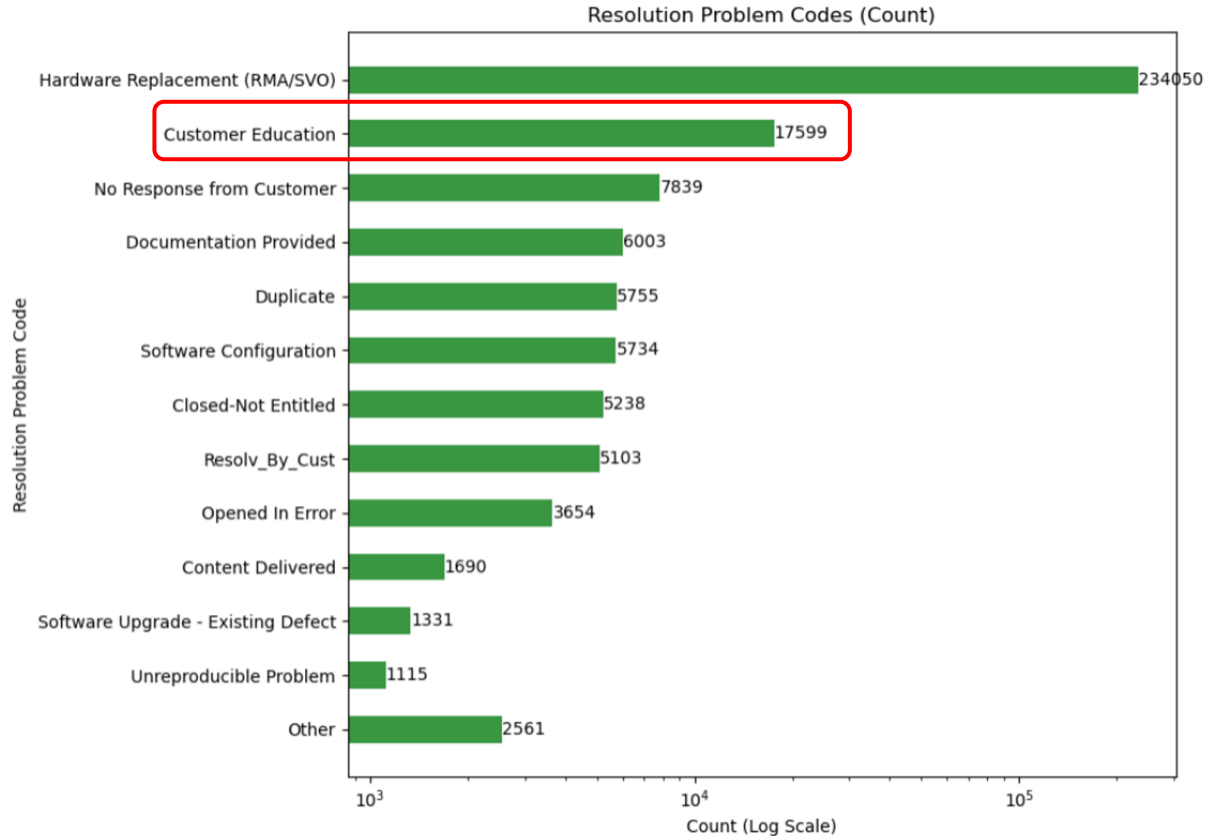
Recommendations

- 1) Promote and incentivize the use of web-based RMA submissions for SMB clients, emphasizing convenience and efficiency over traditional phone channels. This would drive down operational expenses, minimize wait times, and enhance the scalability of request management.
- 2) Implement streamlined software configuration procedures tailored for SMB cases to minimize delays and expedite resolution times.
- 3) Establish better communication channels with automated follow-up systems and multiple options (email, chat, phone) to ensure customers promptly provide necessary information, in order to solve the “No Response from Customer” issue.

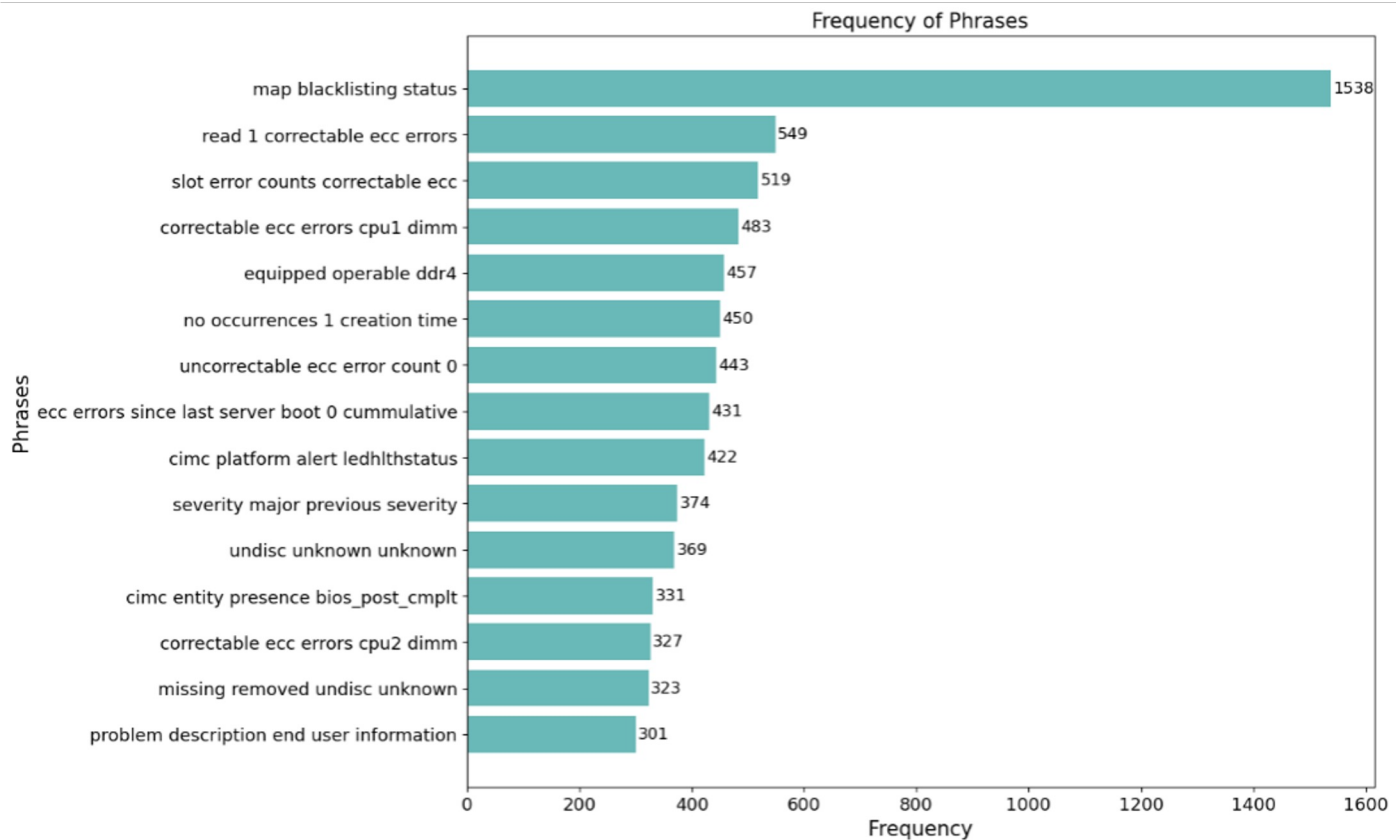
IV. Educational “Resolution”

Customer Education is the second largest problem category.

Cases involving 'Customer Education' rank as the second highest in frequency (17599 cases), hinting at the potential to reduce the cases with revised educational resources (FAQs, detailed manuals, video tutorials, etc).



Common phrases in problem descriptions for Customer Education cases ...



... Instructions for fixing these problems are available online

Automatically send links to FAQs, troubleshooting guides, video tutorials, or manuals when specific common phrases in customer issues are detected.

Mapping Blacklisted or Failed Drive in HX to UCS

 Save  Translations  Download

Updated: June 12, 2019 Document ID: 214508 Bias-Fre

Contents

- Introduction
- Confirming Drive in storfs-support Bundle
- Validate info from HX system
 - Mapping Drive to Host
 - Mapping Drive Serial Number to UCS via UCSM sam_techsupportinfo
- Validate information from UCS system
 - To check via GUI
 - To check via CLI

Introduction

This document walks through mapping a drive which is blacklisted in HX to the drive in UCS. This helps in troubleshooting issues, identifying blacklisted drive and PID of the drive in a Hyperflex environment. We will need both the HX and the UCS logs for this process. Alternatively, you may also run the commands provided on a live system after SSH'ing into the device.

Troubleshooting Correctable ECC Error



CscTsWebDocs Community Member

10-14-2014 07:07 PM - edited

03-01-2019 06:02 AM

[Symptom]

If the fault event shown below occurs on UCSM, you may find from the CIMC log that a correctable ECC error has occurred on DIMM.

Severity: Minor

Code: F0184

Description: DIMM A2 on server X/Y operability: degraded

Name: Memory Unit Degraded

Cause: Equipment Degraded

<https://www.cisco.com/c/en/us/support/docs/hyperconverged-infrastructure/hyperflex-hx-data-platform/214508-mapping-blacklisted-or-failed-drive-in-h.html>

<https://community.cisco.com/t5/data-center-and-cloud-knowledge-base/troubleshooting-correctable-ecc-error/ta-p/3150720>



Recommendations

- 1) **Automated Link Distribution:** Automatically provide immediate responses to inquiries about "map blacklisting," as solutions are accessible online. Based on the analysis of common phrases in problem descriptions related to customer education, develop a system to automatically send educational resources, FAQ links, troubleshooting guides, video tutorials, or detailed manuals when these specific phrases are detected. This approach aims to address customer queries more effectively and reduce repeat cases.
- 2) **Feedback-Driven Content Update:** Use data-driven insights to continuously update and refine educational resources. Analyze customer interactions and feedback on automated links to ensure the content is relevant and effectively reduces the frequency of support cases.

V. RMA Approval

RMA approval for power supply issues should be automated, while RMA approval for Call Home Device reported should be manual.

A mere 7% of RMA requests associated with "Call Home device reported" received RMAs, suggesting a need for further investigation.

automatic rma ucs uecc identified via intersight fault moid

call home device reported

please raise rma

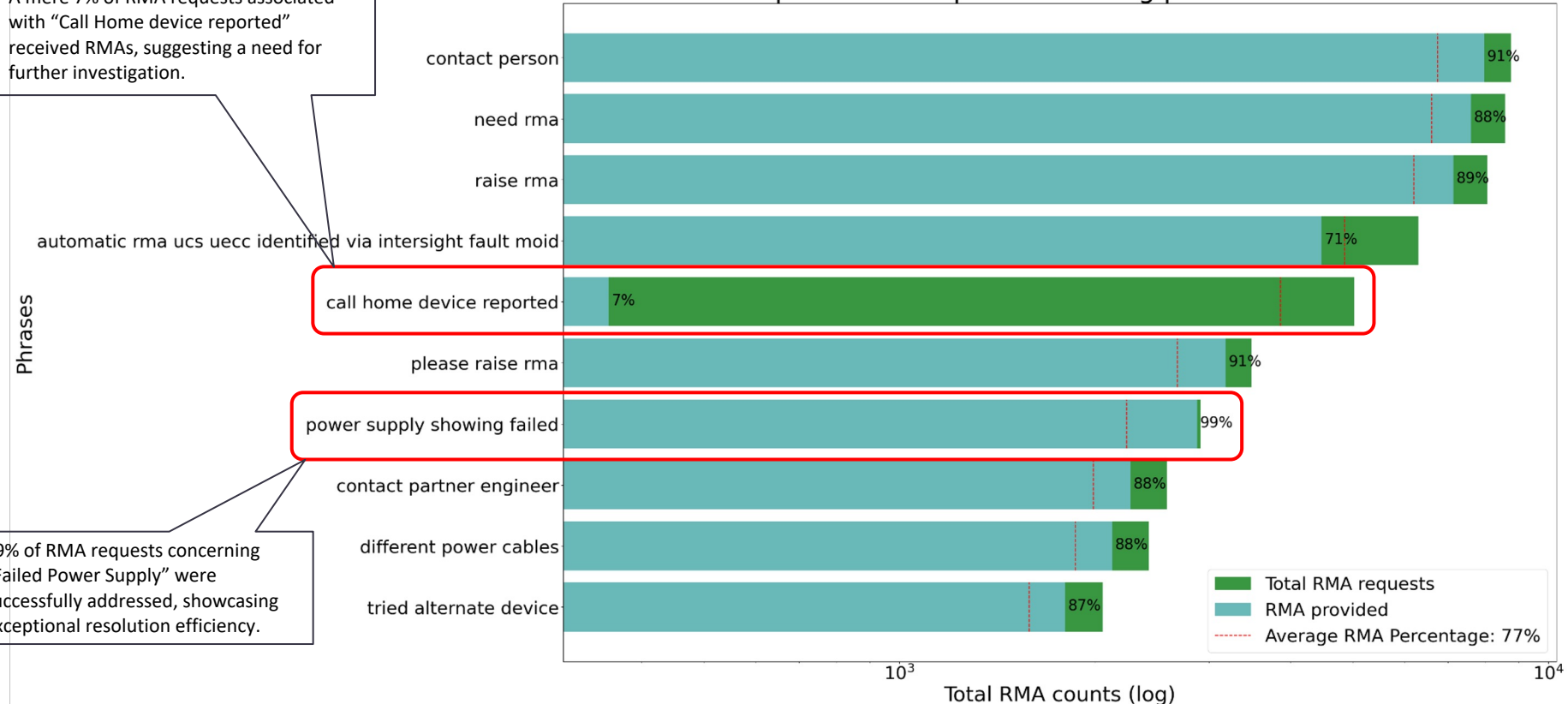
power supply showing failed

contact partner engineer

different power cables

tried alternate device

Total RMA requests for frequent occurring phrases in Problem Details



99% of RMA requests concerning "Failed Power Supply" were successfully addressed, showcasing exceptional resolution efficiency.



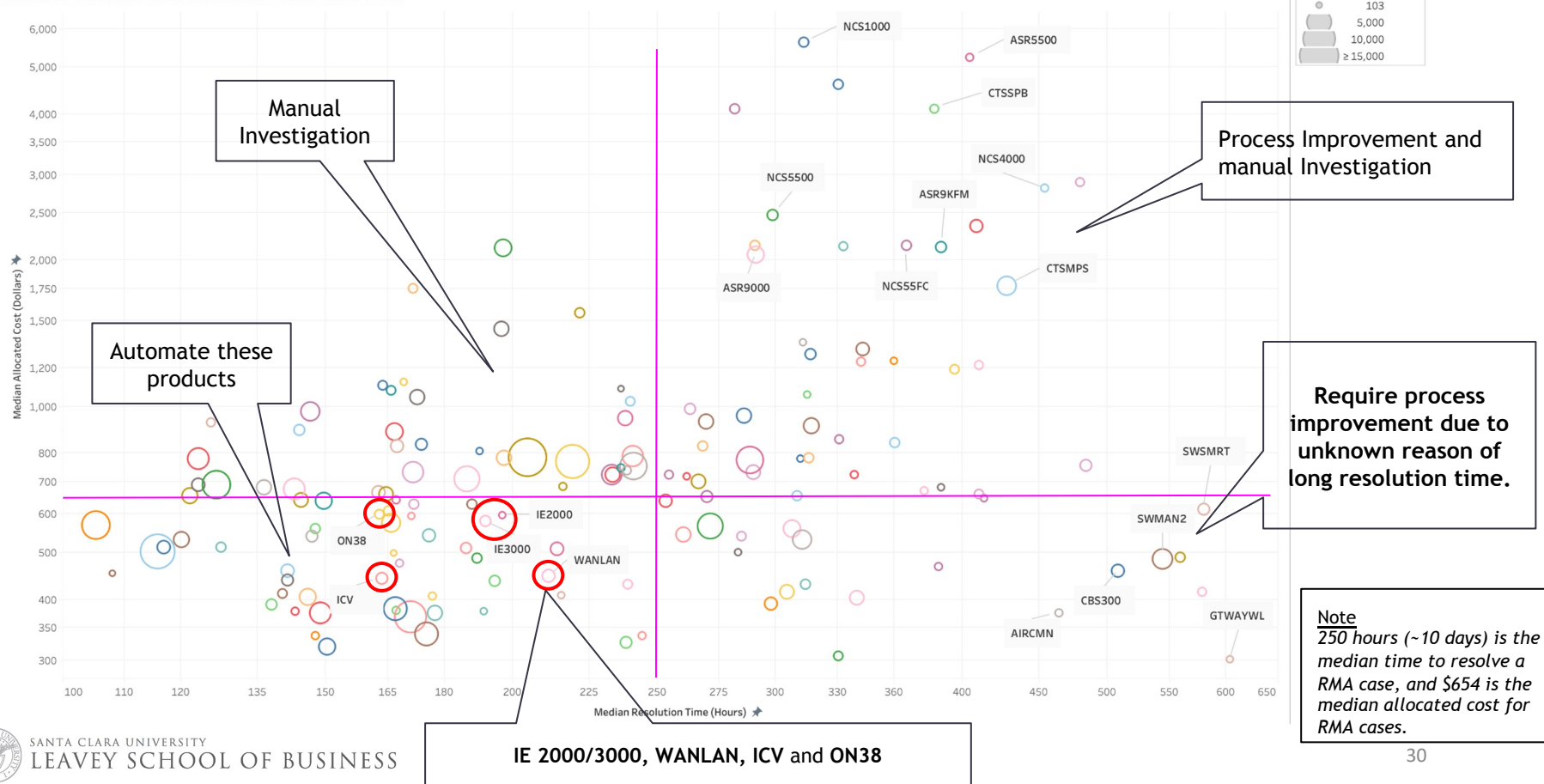
Recommendations

- 1) **Automate RMA Process for Failed Power Supply:** Streamline the RMA process by implementing automated systems tailored to swiftly address cases related to Failed Power Supply. This proactive approach will expedite resolution, ensuring customers receive timely support and minimizing downtime.
- 2) **Thorough Investigation for Call Home Devices:** RMA approval should be manual for Call Home Device reported based on low RMA approval rates.

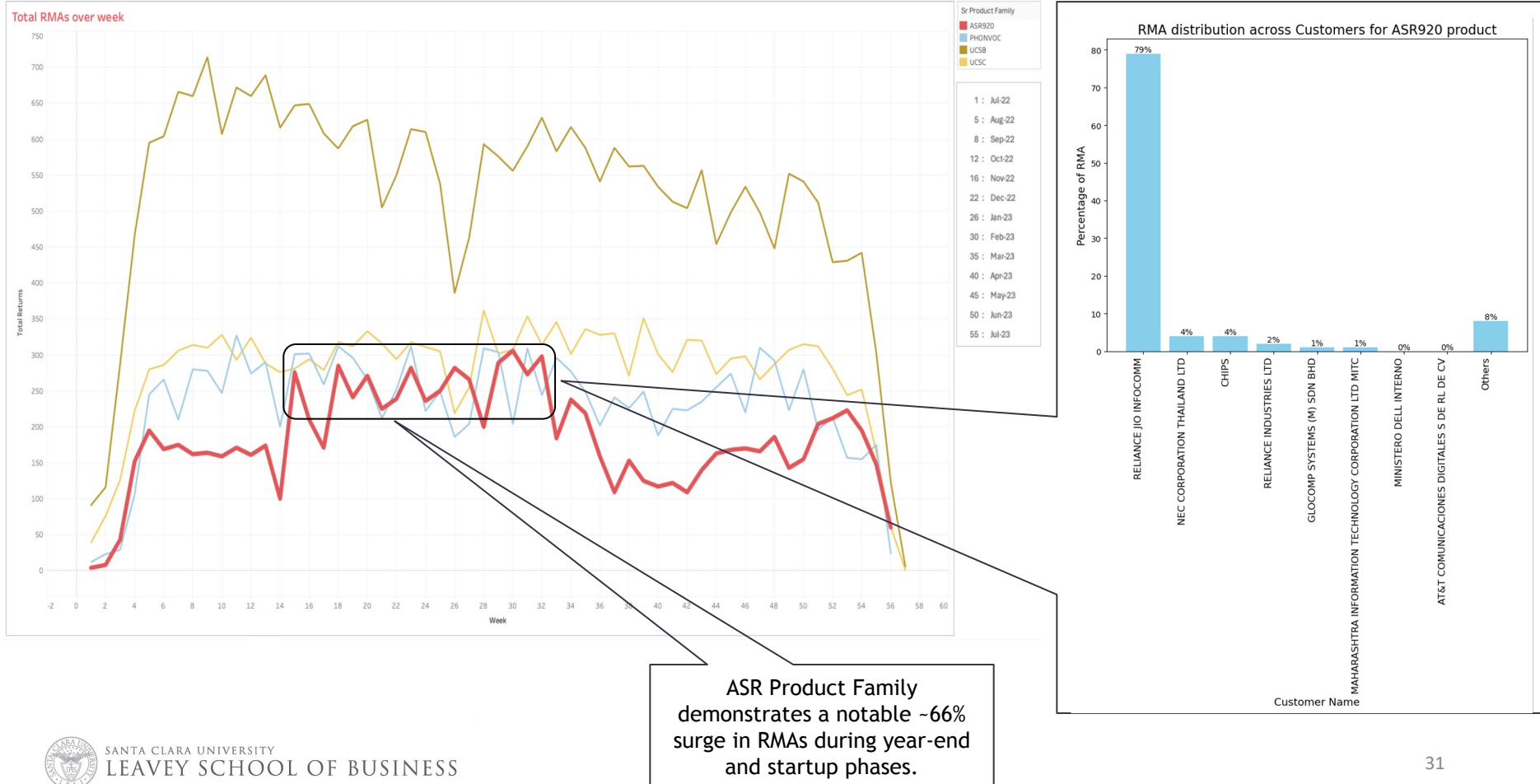
VI. Seasonality and Cost

Considering low median allocated cost and less resolution time, IE 2000/3000, WANLAN, ICV and ON38 should be automated.

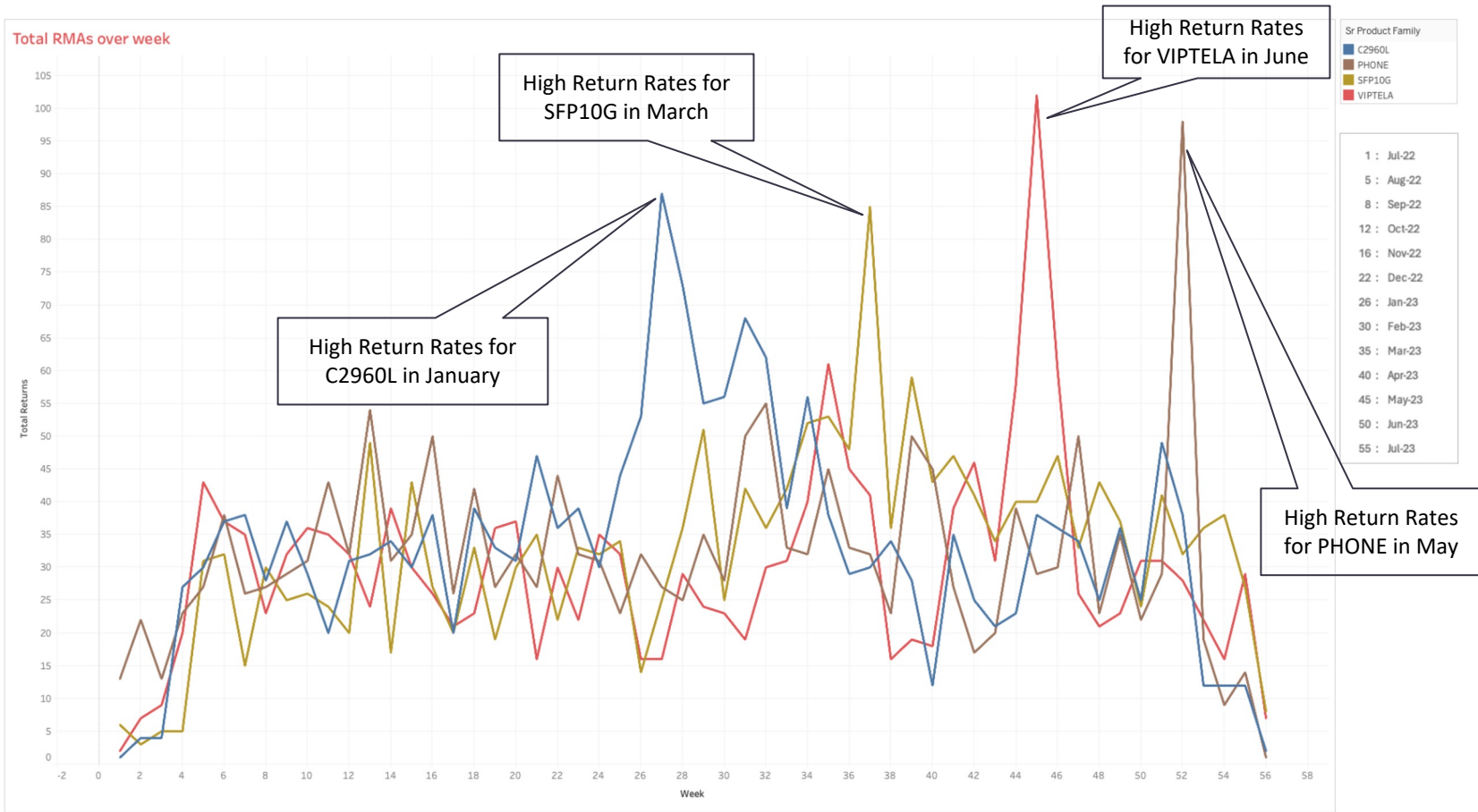
Median Allocated Cost vs. Median Resolution Time



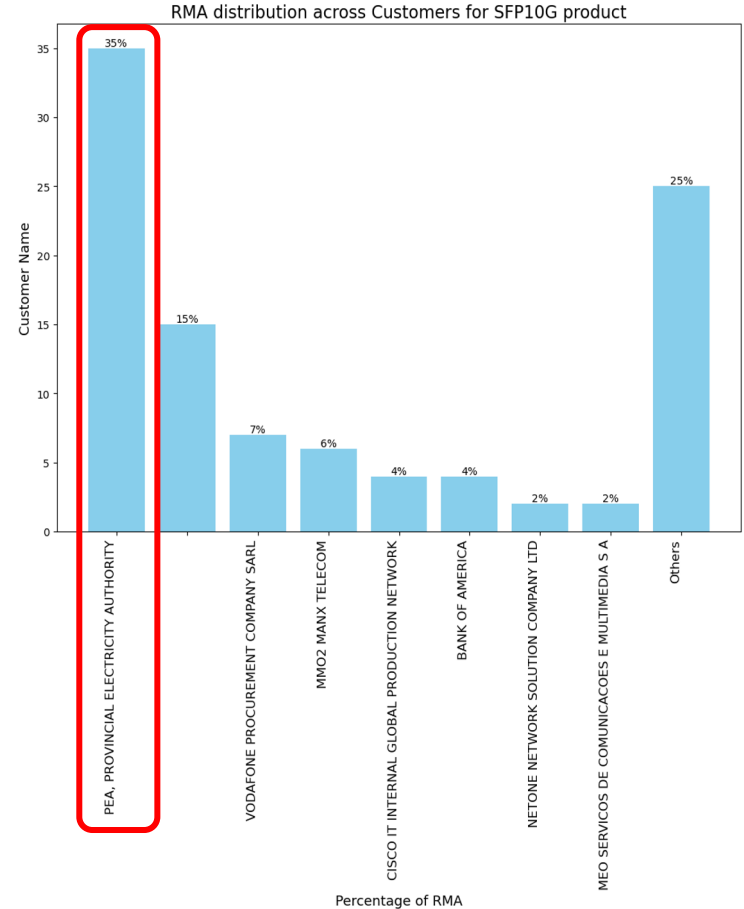
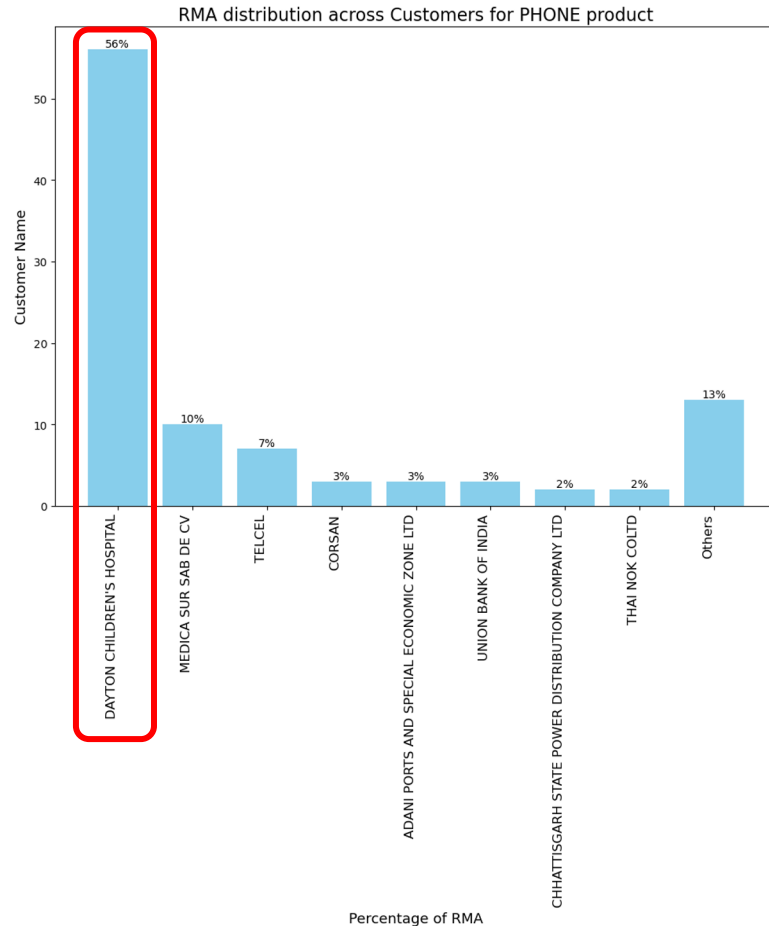
~66% seasonality surge in RMAs in ASR Product Family due mostly to *Reliance Jio*.



Spikes occurred in 4 high volume product families: PHONE, VIPTELA, C2960L, SFP10G ...



... Dayton Children's Hospital and Provincial Electricity Authority are causing the main spikes.



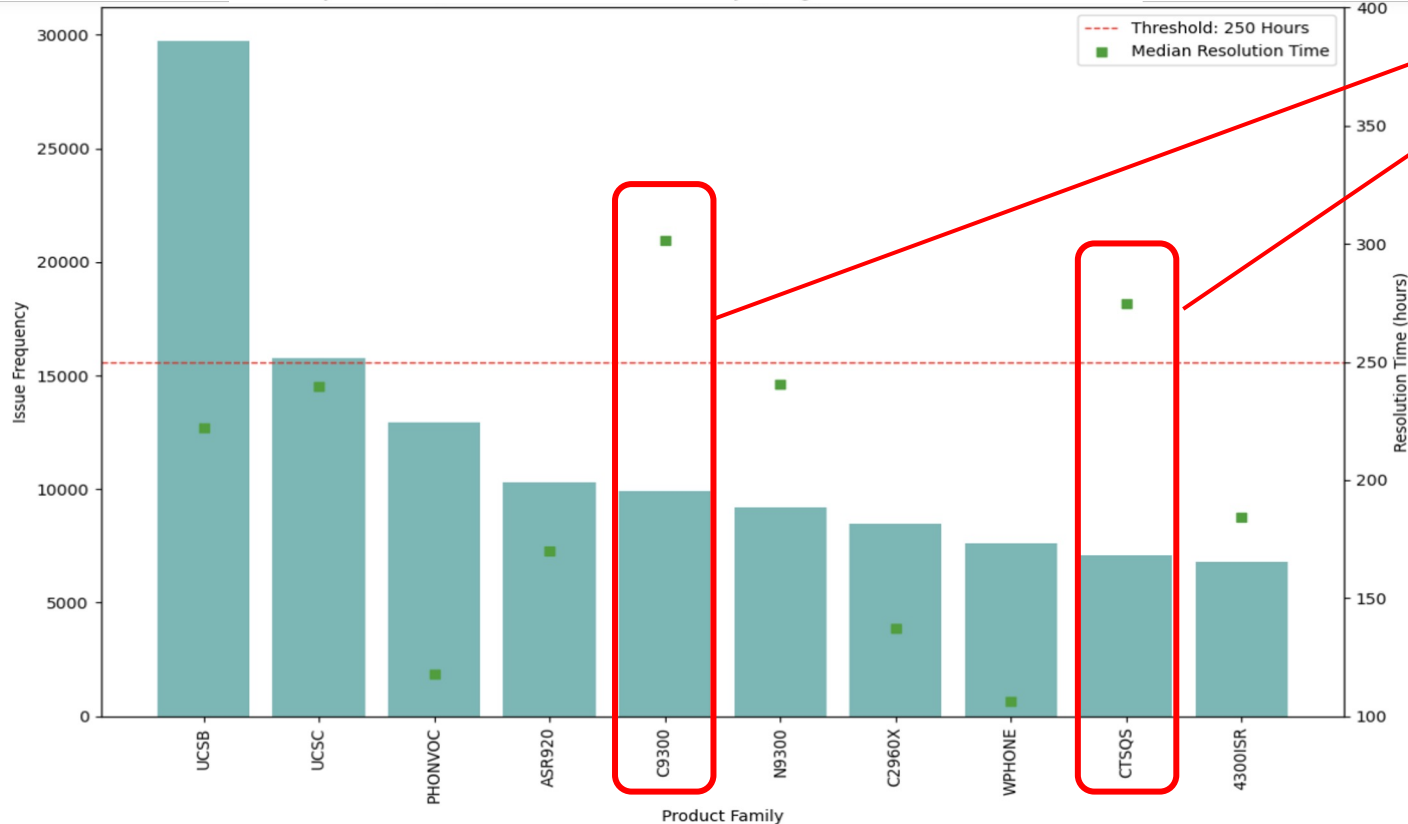
Recommendations

- 1) **Capacity Planning and Resource Allocation:** Enhance capacity planning and resource allocation strategies to effectively manage the increased demand for ASR920 product during year-end and start-up phases, ensuring sufficient inventory of replacement parts and support staff to handle the higher volume efficiently.
- 2) **Collaborative Engagement with Customers:** Foster a collaborative partnership with *Reliance Jio* to identify common pain points and address recurring issues, implementing preventive measures to reduce the frequency of RMA requests originating from their network.
- 3) **Low Cost and High Resolution Time:** Implement process improvement initiatives to streamline workflows, reduce bottlenecks, and enhance efficiency in handling RMA requests, aiming to lower resolution times while maintaining cost-effectiveness.
- 4) **Expensive and High Resolution Time:** Adopt a dual approach of manual intervention and process improvement efforts for cases involving high costs and extended resolution times, combining meticulous manual handling with ongoing process optimization to enhance efficiency and reduce resolution times over time.

VII. Product Families

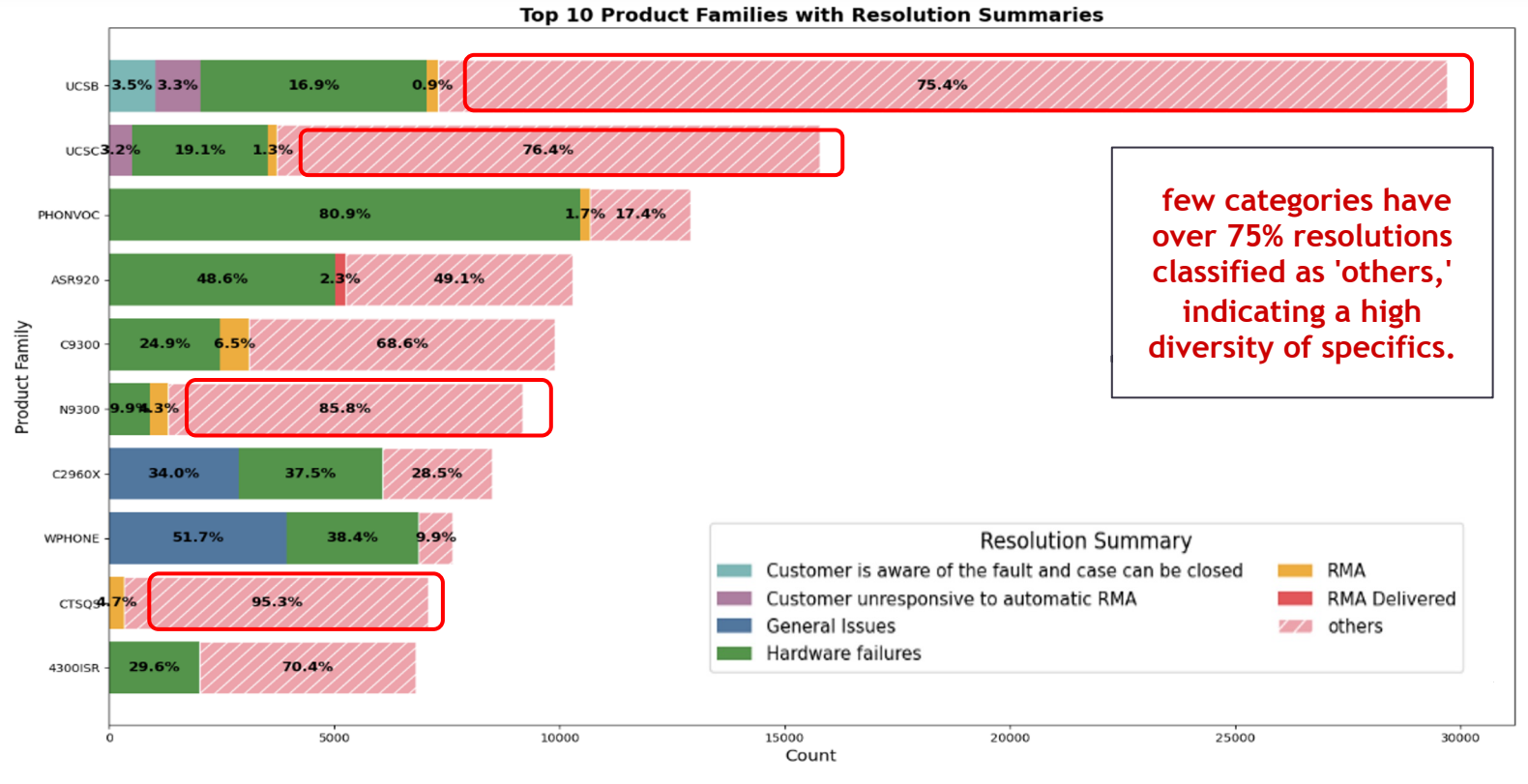
Product Families with fewer reported issues but longer resolution times indicate a need for process improvement.

Top 10 Product Families: Issue Frequency vs. Resolution Time



'C9300' and 'CTSQS' product families have low issue frequency but high resolution time (greater than 250 hours).

Standardizing resolution descriptions would improve issue classification.

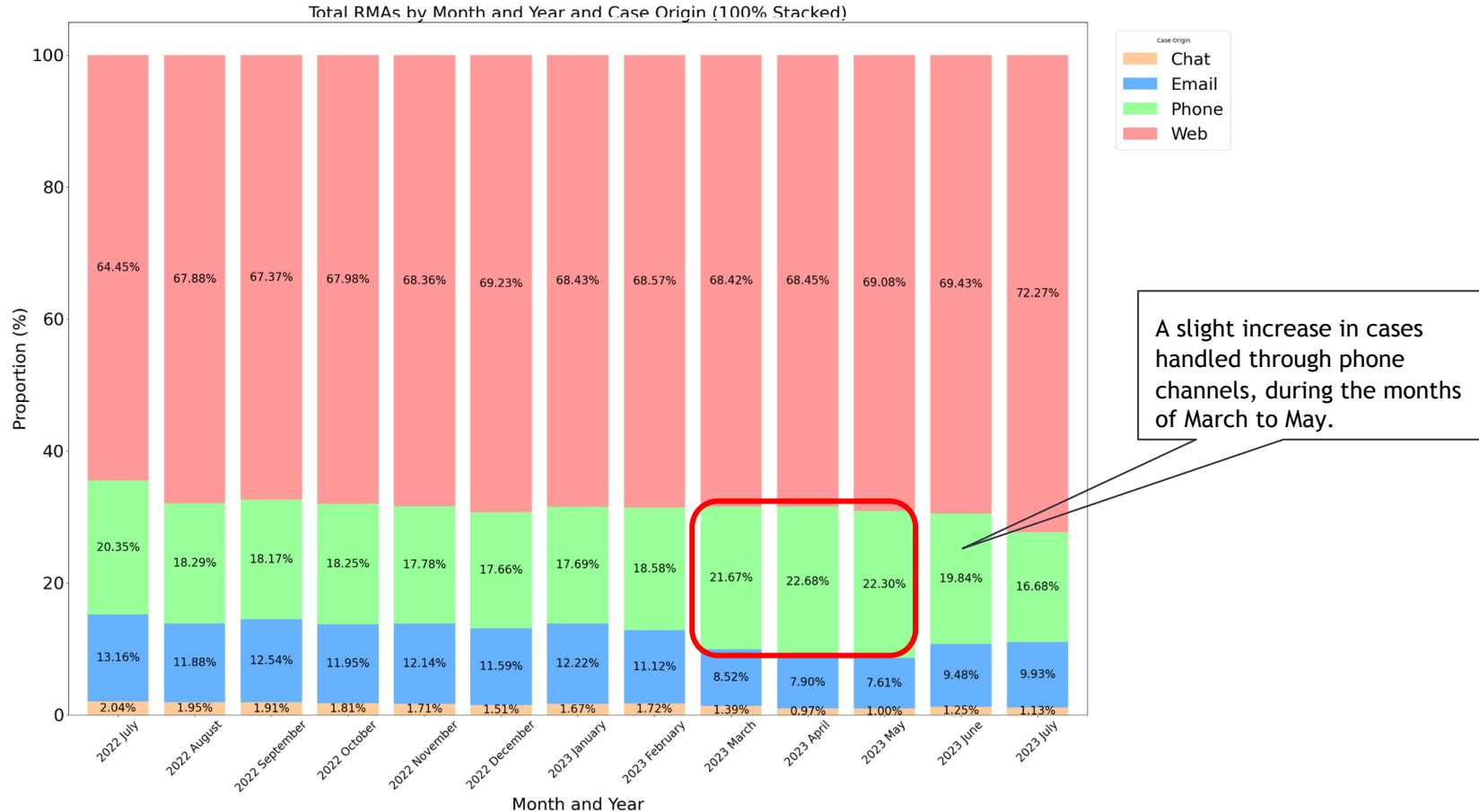


Recommendations

- 1) **Root cause analysis:** Conduct root cause analysis to identify the underlying causes contributing to the extended resolution times.
- 2) **Process improvement initiatives:** Optimize workflows and leverage automation tools to handle repetitive tasks more effectively.
- 3) **Documentation and knowledge base:** Enhance the documentation and knowledge base for common issues within product families to empower support teams to resolve issues more quickly and diminish reliance on specialized expertise.
- 4) **Standardization efforts:** Define a set of standardized categories to align all reported issues. Continuously refine and expand these categories to accommodate emerging issues, ensuring a comprehensive and adaptable classification system.

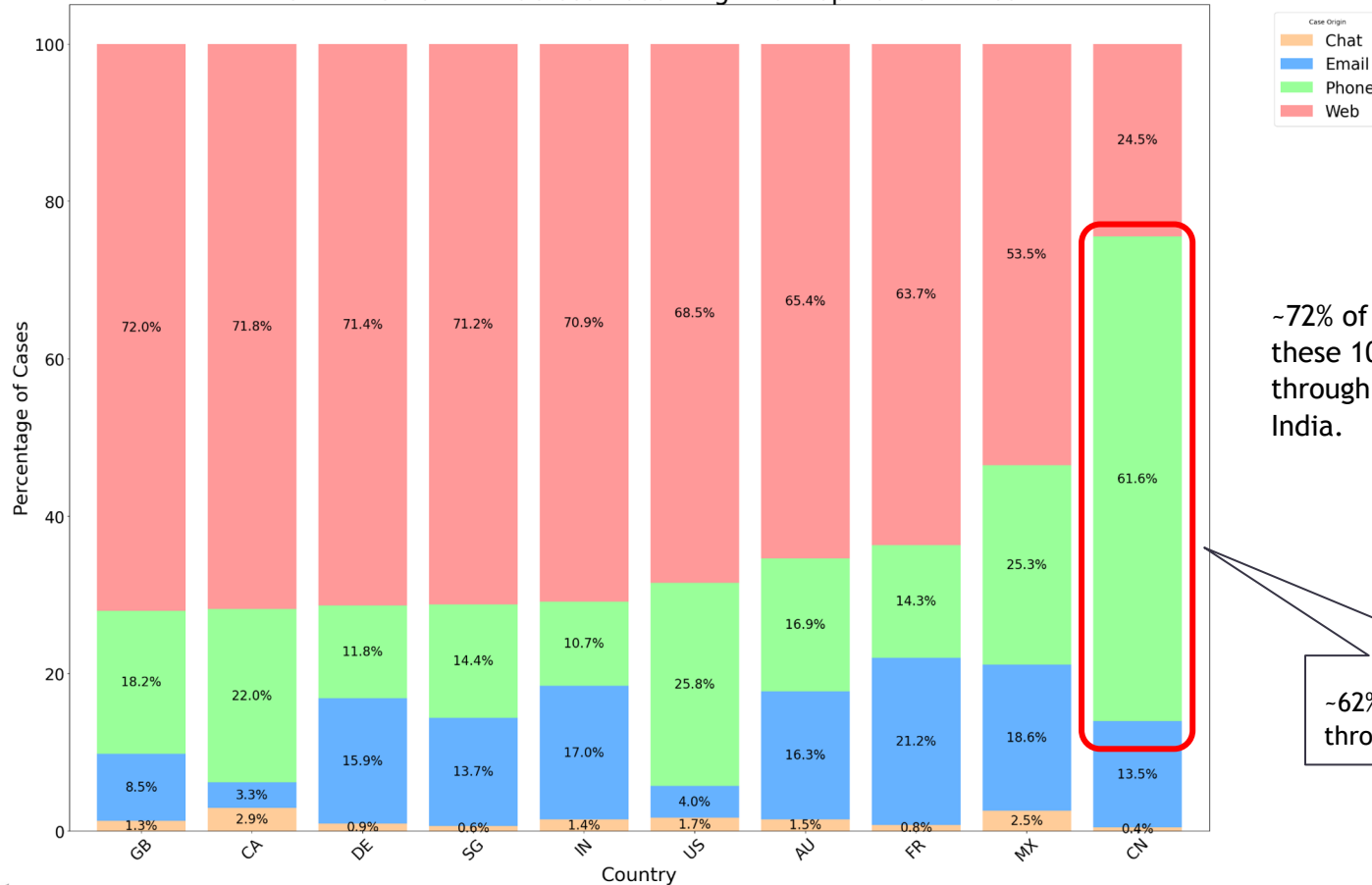
VIII. Case by Channel

March, April, and May show elevated RMA rates for phone support.



Most RMA cases in China come through the Phone channel.

Distribution of RMA across Case Origin for Top 10 Countries



~72% of RMA requests came through these 10 countries, with 41% coming through US, following by 16% through India.

~62% of cases in China came through phone channel.

IX. Next Steps

Next Steps ...

- 1) Enhance web channel efficiency and shift away from phone-based support.
 - a. Implement structured web forms
 - b. Set a target of 90% repair alignment score
- 1) Streamlining SMB channels to low-touch or no-touch interactions.
- 2) Develop multi-language machine learning models to improve issue resolution efficiency, minimize language barriers, and improve customer service experiences.
- 3) Automate distribution of educational resources, enhancing customer self-service capabilities and potentially decreases the number of repeat cases.
- 4) Improve telemetry framework, enabling stronger predictive maintenance capabilities.
- 5) Develop and train an AI-powered chatbot to assist customers with troubleshooting and product inquiries, aiming to minimize false positives and effectively guide customers on how to complete a web form when raising a RMA request.

Questions?

Appendix A - Working Hypotheses



Five High Priority Working Hypotheses

1. Implementing NLP framework could automate more RMA transactions.
 - a. Extract relevant phrases
 - b. Map phrases from text data to different columns (product family, codes etc.)
2. Certain product families may consume more (or less) resolution time.
 - a. Segment data by different product families and compute resolution time
 - b. Identify root causes of resolution time
3. SMB customers consume disproportionate RMA resources.
 - a. Classify customers by customer types
 - b. Review RMA automation levels

Five High Priority Working Hypotheses (Cont'd)

4. Resolution time is dependent on the channel through which an item enters the RMA process.
 - a. Develop resolution time by product family by channel
 - b. Rationalize product families by channel based on product cost and complexity
5. RMA Data can be used to predict Upsell Opportunities.
 - a. Segment the frequency of RMAs by customer; identify customers with recurrent product problems such as common reasons for returns
 - b. Look for patterns that indicate a higher need for maintenance services or replacement
 - c. Review service contracts to possibly reduce RMA frequency and enhance customer satisfaction
 - d. Risk mitigation

Five Lower Priority Working Hypotheses

6. RMAs with high allocated cost and high-resolution time may indicate inventory issues.
 - b. Correlate inventory issues with high cost /high allocated time items
 - c. Identify how to improve forward inventory planning for out-of-stock items
7. Seasonality disproportionately affects selected product families.
 - a. Identify RMA peaks by product family
 - b. Review failure rates with peak periods to identify root causes

Five Lower Priority Working Hypotheses (Cont'd)

8. High RMA data input quality and completeness can provide more accurate failure classification leading to more efficient allocation of resources.
 - a. Evaluate the quality of initial data inputs from both Web and Phone channels, as they collectively account for 85% of the total cases in the previous dataset
 - b. Enhance the metadata tagging quality

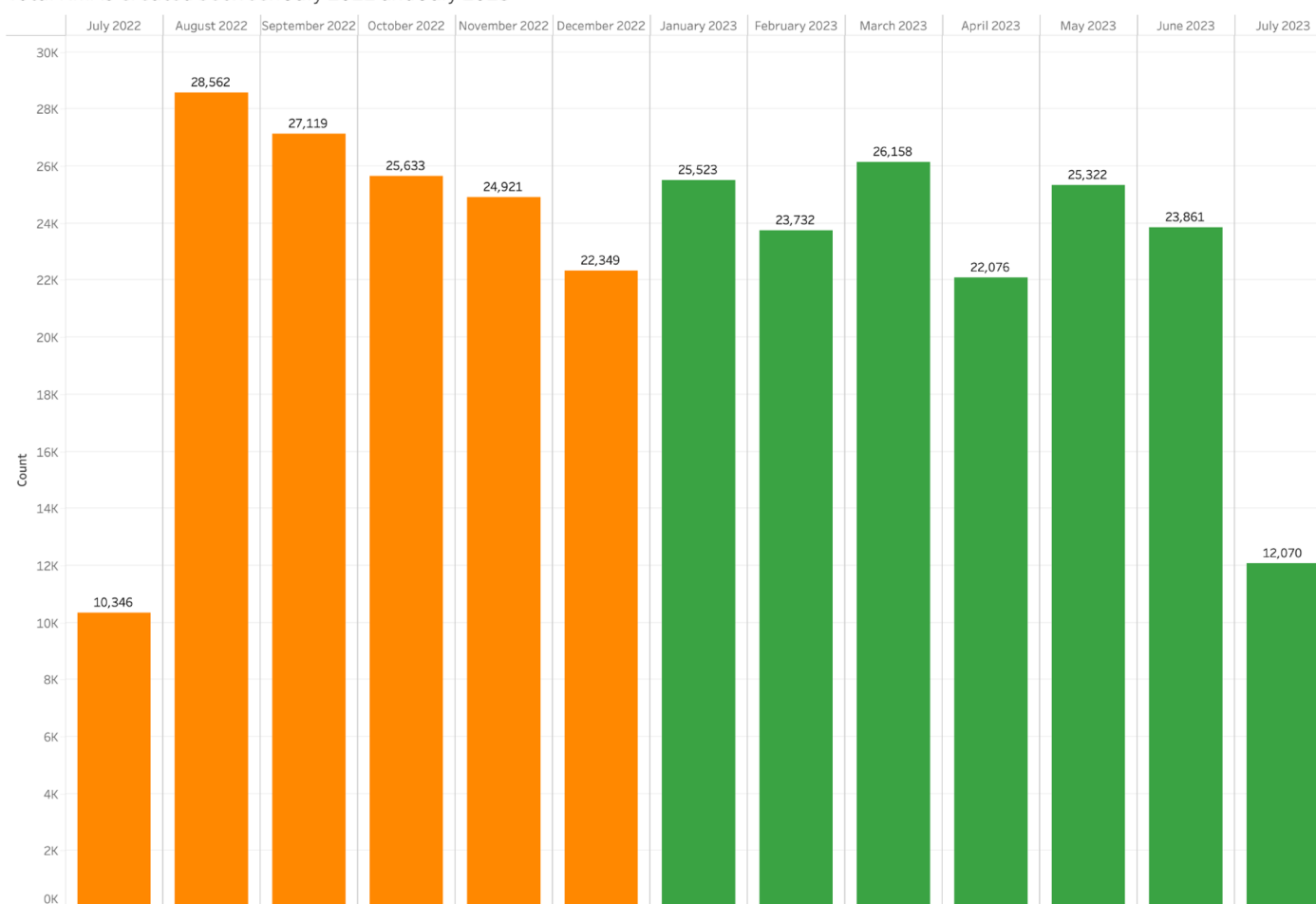
Five Lower Priority Working Hypotheses (Cont'd)

9. High RMA cost relative to total product value could imply product quality issues or operational inefficiencies.
 - a. Examine the influence of usage patterns and environmental conditions on high-value product performance through geospatial analysis using external geographic data
 - b. Evaluate the logistics operations for factors that may contribute to higher cost and possible suboptimal resource allocation
10. Certain product families have a significant number of repair problems that can be solved by education, software or hardware replacement.
 - a. Analyze the problem data to identify keywords - software or hardware
 - b. The revision of review/develop policy for certain product families
Incorporate targeted educational content

Appendix B - Dataset Distribution



Total RMAs Created between July 2022 and July 2023



YEAR(Month)
■ 2022
■ 2023

A total of
297,672 cases
 were created in
 the period from
 July 2022 to July
 2023.

**July data is incomplete causing
 lower count.*

Thank You

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